

Questions and Answers with Professor Emeritus Paul A. Zeitz, Class of 1975



Paul A. Zeitz, Class of 1975, is a Professor Emeritus of Mathematics at the University of San Francisco, where he also served as Chair of the Mathematics Department. He is also the author of the classic text, The Art and Craft of Problem Solving (1998).

At Stuyvesant, Zeitz served as captain of the Math Team. At age 16, he qualified for the very first U.S. team to participate in the International Mathematical Olympiad (IMO) in 1974, where he won a bronze medal. He later contributed to writing national math contest problems, including USAMO. He would eventually go on to coach multiple U.S. IMO teams, including the first ever perfect-scoring 1994 “Dream Team.”

He received his bachelor’s degree from Harvard University in 1981, having won a Westinghouse (now Regeneron) Science Talent Search scholarship. He earned his Ph.D. from the University of California, Berkeley in 1992.

In 1999, Zeitz co-founded the Bay Area Mathematical Olympiad and co-founded the San Francisco Math Circle in 2005. In 2009, he produced a series of video lectures for The Teaching Company on problem solving. The Mathematical Association of America awarded Zeitz with the Deborah and Franklin Haimo Awards for Distinguished College or University Teaching of Mathematics in 2003.

Currently, Zeitz serves as Chair of the Board of Directors of The Proof School, located in downtown San Francisco. Starting in 2013, Zeitz envisioned and then co-founded this mathematics-focused middle school and high school which opened its doors in 2015 and features a unique and innovative world-class curriculum.

Paul Zeitz's answers are taken from an interview conducted by Josephine Lee and have been edited for clarity.

1) In 2011, you wrote the foreword to *Math Survey*, “The Three Epigraphs,” to encourage students to get lost, get obsessed, and get together. From your experience, what are the most common stumbling blocks for young explorers, and what is your recommendation to overcome them?

The most common stumbling blocks are psychological; almost every beginner lacks confidence and gives up too easily. Most beginners do not spend enough time thinking about problems and have been conditioned to think that if they don't get “the answer” within a minute or two, then they never will. In reality, most worthwhile things that one grapples with require large amounts of time, failure, wild goose chases, etc. And a baseball-style average (say, 1 in 3 successes) is very, very good.

So the recommendation to deal with this is:

- Get used to a long time scale (working for an hour, a day, etc.)
- When you fail to solve a problem, don't give up but perhaps set it aside
- Simultaneously, improve your confidence by doing problems that are easy for you, at least some of the time
- Find trusted comrades that you can work with whom you don't feel (too much) competition.

2) What do you remember most about your experience at Stuyvesant? Were there any specific students, teachers, or extracurricular activities of Stuyvesant that have had a significant impact on you?

I loved Stuyvesant. It was safe, and everyone was fun to talk to. I am still close friends after nearly 50 years with a few fellow students. Of course, I have very fond memories of Math Team, which back then was just a couple dozen people, completely run by students (although our faculty advisor was a fine mathematician who was very kind to us, but she let us run the show). The older students mentored the younger students. I had the good fortune to be mentored by Eric Lander, who became a famous scientist and President Biden's cabinet-level science advisor.

My very best friends were not Math Team people. I spent lots of time taking very long walks all over NYC with my friends, playing frisbee, hanging out in parks, listening to concerts for free at Lincoln Center by discovering noise leaks in the parking garage, etc. My favorite teachers were Sylvia Schwartz, who taught math with a tremendous aesthetic sense (she was a serious violinist) and Rlene Dahlberg (not misspelled), who taught an independent study English class that was very stimulating, and Vivianne Gluck, who taught history. She inspired me to read history seriously and indeed I became a history major in college (American History with a minor in East Asian History).

3) In your experience as a mathematician and educator, what is one thing you've changed your mind about in the last decade?

I taught high school in an independent private high school in Colorado after college. I discovered that, contrary to my expectations, one of my favorite classes was a terminal algebra class for “weak” math students. The students were pretty uniform in their skill level (which was low), and they considered themselves to be “dumb.” Their expectations were quite low, so it turned out to be fun to fool them into doing interesting and hard math through puzzles. They were a very cohesive group that helped one another, and they (and I) had tons of fun.

At the other end of the spectrum, I also loved training the IMO teams in the mid-1990s, including the 1994 “Dream Team” that got a perfect score (no team matched this achievement until China in 2022). These students were super-skilled and super talented, and I had to work like a maniac to keep up with them. So I never got much sleep, but we had tons of fun. One thing that I discovered is that no matter how “smart” you are, you often feel “stupid,” because there are always problems that one cannot solve. A large part of my job was helping these students to learn to accept their humanity and relax when something was (perhaps temporarily) elusive to them.

4) Is there an important truth that you find many/most others disagree with? If so, what is it?

Many people, especially at places like Stuyvesant, confuse “intelligence” with virtue. In reality, intelligence is just an attribute. Being “smart” has, sadly, nothing to do with being “good.” A “good” student is not necessarily a “smart” or a “accomplished” student.

5) Is there anything else you would like to communicate to current Stuyvesant students?

Every year, it seems to get harder to get into college. It is easy to get overwhelmed and anxious, especially with nervous parents hovering over you. Please relax. First of all, even if you get into your “dream” college, it is just the minor step in a very long path. And if you don't get in, your life is NOT over. They made a mistake, you'll get over it, and you will have a happy and successful life if you let yourself have one. You are the one in control. Strive for situations that give you as much autonomy as possible. This is an undervalued quality that is, in my opinion, the key to a good life.